DAY 11 | HANDS-ON ACTIVITIES

React Basics: Problem Statements

Seven problems. In Colab, use show_react() from the notebook. In VS Code, use a Vite React project and edit src/App.jsx.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Hello Component

Level: Warm-up | **Concepts:** component, JSX expressions

Create a Greeting component that shows your name, today's date and a message that says "Good morning" or "Good evening" based on the current hour.

Sample run / expected result

```
Hello, Aiko!
Today is Mon Nov 09 2026
Good morning
```

Hints

- new Date().getHours() < 12

Problem 2: Profile Cards

Level: Easy | **Concepts:** props, reuse

Create ProfileCard with props name, role and city. Render three cards for three different people.

Sample run / expected result

```
[Mei | Data Engineer | Tokyo]
[Arjun | FDE | Bengaluru]
[Ken | Designer | Osaka]
```

Hints

- function ProfileCard({ name, role, city })

Problem 3: Counter with Limits

Level: Easy | **Concepts:** useState, onClick

Build a counter with +1, -1 and Reset buttons. The count must never go below 0 or above 10; disable buttons at the limits.

Sample run / expected result

```
Count: 10   [+1 disabled] [-1] [Reset]
```

Hints

Problem 4: Live Converter

Level: Medium | **Concepts:** controlled input, derived values

Build a Celsius input that shows Fahrenheit and Kelvin as you type. Show "Enter a number" when the input is empty or not a number.

Sample run / expected result

```
Celsius: 37
Fahrenheit: 98.6 | Kelvin: 310.15
```

Hints

- Number(value) and Number.isNaN()

Problem 5: Layout with children

Level: Medium | **Concepts:** children prop, composition

Create a Panel component with a title prop and children. Build a page with three panels: About, Skills and Contact, each with different content.

Sample run / expected result

```
Three styled panels, each with a heading and custom content
```

Hints

- function Panel({ title, children })

Problem 6: Like Button

Level: Challenging | **Concepts:** boolean state, conditional text, styles

Build a like button that toggles liked state, changes its label and colour, and a separate total likes counter that goes up and down accordingly.

Sample run / expected result

```
[Liked] 13 likes  ->  click  ->  [Like] 12 likes
```

Hints

- setLiked(!liked); setLikes(l => liked ? l - 1 : l + 1)

Problem 7: Mini Signup Form

Level: Stretch | **Concepts:** multiple controlled inputs, onSubmit, validation

Build a form with name, email and password. Show validation messages as the user types (email must contain @, password at least 8 characters). On submit, show a welcome message instead of the form.

Sample run / expected result

```
Password must be at least 8 characters
... after submit: Welcome, Mei!
```

Hints

- Use one state object: const [form, setForm] = useState({ name: "", email: "", password: "" })
- e.preventDefault() in onSubmit

Ganbatte (頑張って): do your best